1. The number of unpaired electrons in Cr+ will be

(a) 3
(b) 4
(c) 5
(d) 6

Answer: (c) 5

Explanation: Cr = [Ar] 3d5 4s1. Removing one 4s electron gives Cr+ = [Ar] 3d5, so it has 5 unpaired electrons.

2. The highest oxidation state of Cr will be

(a) 2
(b) 3
(c) 4
(d) 6

Answer: (d) 6

Explanation: Chromium reaches a maximum oxidation state of +6, for example in chromate/dichromate compounds.

3. Which statement is true about the transitional elements

(a) They are highly reactive
(b) They show variable oxidation states
(c) They have low M.P.
(d) They are highly electropositive

Answer: (b) They show variable oxidation states

Explanation: The correct choice is They show variable oxidation states. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

4. The transitional metal which form green compound in +3 oxidation state and yellow orange compound in +6 oxidation state is

(a) Fe
(b) Ni
(c) Cr
(d) Co

Answer: (c) Cr

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.

5. Highest (+7) oxidation state is shown by

(a) Co
(b) Cr
(c) V
(d) Mn

Answer: (d) Mn

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.

6. Transitional elements are

(a) All metals
(b) Few metals and few non-metals
(c) All solids

(d) All highly reactive

Answer: (a) All metals

Explanation: The correct choice is All metals. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

7. Which of the following has highest ionic radii

(a) Cr+3

(b) Mn+3

(c) Fe+3

(d) Co+3

Answer: (a) Cr+3

Explanation: The correct choice is Cr+3. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

8. In a reaction the ferrous (Fe++) iron is oxidised to ferric (Fe+++) ion. The equivalent weight of the ion in the above reaction is equal to

(a) Half of the atomic weight

(b) 1/5 of the atomic weight

(c) The atomic weight

(d) Twice the atomic weight

Answer: (c) The atomic weight

Explanation: Fe2+ -> Fe3+ + e-. One electron is transferred, so equivalent weight = atomic weight/1 = atomic weight.

9. Which of the following element has maximum density

(a) Hg

(b) Au

(c) Os

(d) Pb

Answer: (c) Os

Explanation: Compare the relevant atomic masses and/or densities; the listed correct option has the greatest value among the choices.

10. Which is heaviest among the following

(a) Iron

(b) Copper

(c) Gold

(d) Silver

Answer: (c) Gold

Explanation: Compare the relevant atomic masses and/or densities; the listed correct option has the greatest value among the choices.

11. Transitional elements exhibit variable valencies because they release electrons from the following orbits

(a) ns orbit

(b) ns and np orbits

(c) (n – 1)d and ns orbits

(d) (n – 1)d orbit

Answer: (c) (n – 1)d and ns orbits

Explanation: The correct choice is (n – 1)d and ns orbits. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

12. The tendency towards complex formation is maximum in

(a) s - block elements
(b) p - block elements
(c) d - block elements
(d) f - block elements

Answer: (c) d - block elements

Explanation: Transition-metal ions form complexes readily because of their relatively high charge density and available orbitals for accepting ligand electron pairs.

13. Which forms coloured

(a) Metals
(b) Non-metals
(c) p - block elements
(d) Transitional elements

Answer: (d) Transitional elements

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

14. Which element belongs to d – block

(a) Na
(b) Ca
(c) Cu
(d) Ar

Answer: (c) Cu

Explanation: The correct choice is Cu. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

15. Variable valency is shown by

(a) Na
(b) Cu
(c) Mg
(d) Al

Answer: (b) Cu

Explanation: Transition elements can use both ns and (n-1)d electrons in bonding because these levels are close in energy, so several oxidation states are possible.

16. The element with a atomic number 26 is

(a) A non-metal
(b) Krypton
(c) Iron
(d) Manganese

Answer: (c) Iron

Explanation: The correct choice is Iron. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

17. One of the following metals forms a volatile carbonyl compound and this property is taken advantage of for its extraction. This metal is

(a) Iron
(b) Nickel
(c) Cobalt
(d) Tungston

Answer: (b) Nickel

Explanation: Nickel forms volatile Ni(CO)4 in the Mond process, which is used for purification/extraction of nickel.

18. The coinage metals are

(a) Iron, Cobalt, Nickel
(b) Copper and Zinc
(c) Copper, Silver and Gold
(d) Gold and Platinum

Answer: (c) Copper, Silver and Gold

Explanation: The traditional coinage metals are Cu, Ag and Au.

19. Which of the following structure is that of a coinage metal

(a) 2, 8, 1
(b) 2, 8, 18, 1
(c) 2, 8, 8
(d) 2, 18, 8, 3

Answer: (b) 2, 8, 18, 1

Explanation: The correct choice is 2, 8, 18, 1. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

20. An elements in+3 oxidation state has the electronic configuration (Ar)3d3. Its atomic number is

(a) 24
(b) 23
(c) 22
(d) 21

Answer: (a) 24

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.